

# Docker Installation

## For Windows

### 1. Install WSL2

以系統管理員身份開啟 PowerShell，執行

```
wsl --install -d Ubuntu-22.04
```

安裝完成後重新開機。

(使用其他版本 Ubuntu 也可以，如果沒有辦法裝 WSL2 用 WSL 替代)

確認 WSL 是否安裝成功：

```
wsl -l -v
```

若看到 Ubuntu 並且 VERSION 為 2 即代表成功。

### 2. Launch Ubuntu

第一次啟動 Ubuntu 時：

- 建立 Linux 使用者名稱
- 建立密碼

更新套件：

```
sudo apt update
```

### 3. Install Docker Desktop

下載 Docker Desktop：

<https://docs.docker.com/desktop/setup/install/windows-install/>

安裝時，勾選「Use WSL2 instead of Hyper-V」

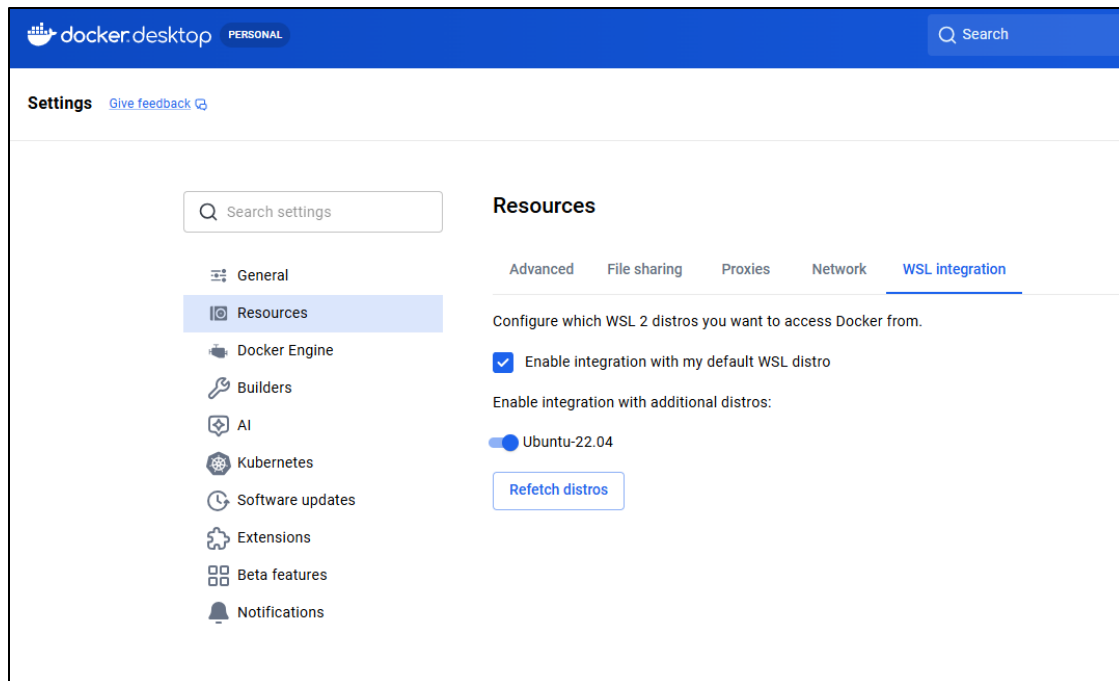
安裝完成後重新開機或登出。

### 4. Enable WSL Integration

打開 Docker Desktop：

Settings → Resources → WSL Integration

將 Ubuntu-22.04 打開 Enable integration.



## 5. Test Docker

在 WSL 或是 Powershell 終端機輸入：

```
docker --version
```

測試 container：

```
docker run hello-world
```

若成功看到 hello-world 訊息代表 Docker 正常。

```
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:0e760fd9bc48ba8041e7c6db999bb40bfca508b4be580ac75d32c4e29d202ce1
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/
```

**Note:** 記得每次要用之前先啟動 Docker Desktop .

## For Ubuntu (root)

You can check the [Link](#) for other Linux Distributions

### Quick Install Script

```
curl -fsSL https://get.docker.com -o get-docker.sh
sudo sh get-docker.sh
sudo service docker start
sudo docker run hello-world
```

## For Ubuntu (rootless)

You can check the [Link](#) for more information.

(使用 GPU 不建議)

### Quick Install Script

```
curl -fsSL https://get.docker.com/rootless | sh
docker run hello-world
```

### Missing uidmap Error

若看到以下錯誤

[ERROR] Missing system requirements. Run the following commands to  
[ERROR] install the requirements and run this tool again.

表示要先安裝 uidmap

```
sudo apt install -y uidmap
```

## Docker Tutorial

根據 Docker 的安裝方式與權限設定，部分環境可能需要在 docker 指令前加上 sudo。

### 1. Create a Container

```
docker run -it --name <container_name> <image_name>
```

### 2. Show All Containers

```
docker ps -a
```

### 3. Exit a Container

```
exit
```

### 4. Show Downloaded Images

```
docker images
```

### 5. Restart a Container

```
docker start -ai <container_name>
```

### 6. Enter a Running Container

(新開一個 shell)

```
docker exec -it <container_name> bash
```

### 7. Stop a Container

```
docker stop <container_name>
```

### 8. Remove a Container

```
docker rm <container_name>
```

### 9. Volume Mount

```
docker run -it --name <container_name> \  
-v <host_path>:<container_path> <image_name>
```

### 10. Copy Files from Host to Container

```
docker cp <host_path> <container_name>:<container_path>
```

### 11. Copy Files from Container to Host

```
docker cp <container_name>:<container_path> <host_path>
```

## CUDA / NVIDIA Driver Notes

### Driver Check

Ensure that your NVIDIA driver supports **CUDA Version >= 12.4**.

You may check this using:

```
nvidia-smi
```

If not, please install a newer NVIDIA driver.

### Nsight Compute (ncu) Permission Notes

#### Windows

NVIDIA 控制面板

→ Desktop → Enable Developer Settings

→ Developer → Manage GPU Performance Counters

→ Allow access to the GPU performance counter to all users



#### Linux

```
echo "options nvidia NVreg_RestrictProfilingToAdminUsers=0" | \
sudo tee /etc/modprobe.d/nvidia-prof.conf
sudo update-initramfs -u
sudo reboot
```

重新開機後，確認設定是否成功：

```
cat /proc/driver/nvidia/params | grep RestrictProfiling
```

若看到 RestrictProfilingToAdminUsers: 0 代表已成功開啟 GPU profiling 權限。